Relation Algebra

§1. Notation

Traditionaly Adjunction F -| G is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$F(x) \leq F(x) \implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x)))$$
$$G(y) \leq G(y) \implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y)$$

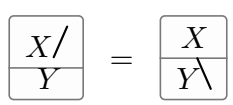
By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

(1a)

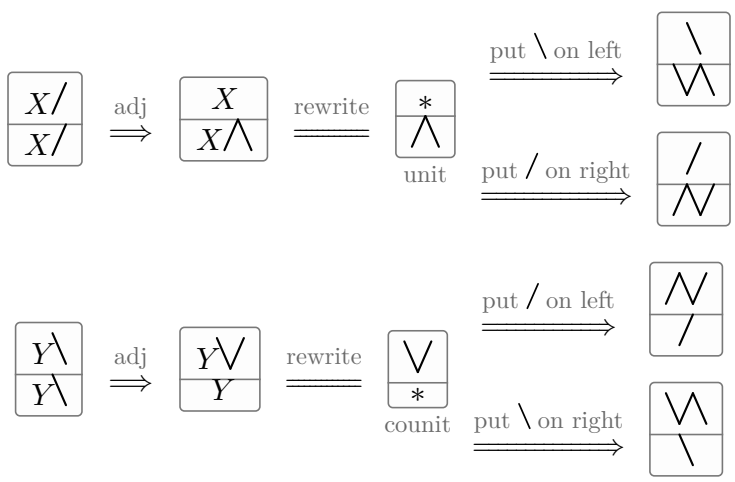
1 • write F as $/$, G as $\backslash$

2 • render $a \leq b$ as $\frac{a}{b}$

3 • **rewrite** $X \leq XF$ as $1 \leq F$



adj



now we have $\diagup\!\!\diagdown = /$ and $\diagdown\!\!\diagup = \backslash$, and you just discovered string diagrams.

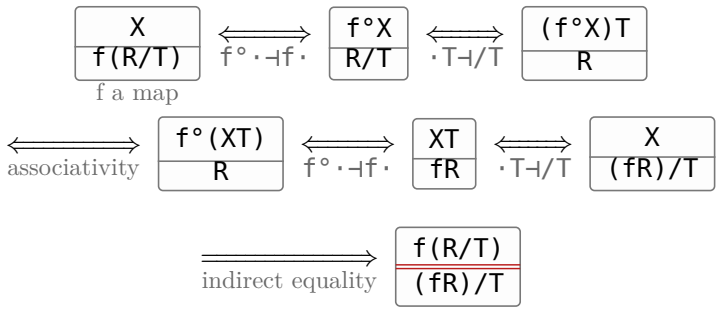
§1.1. Adjunctions

(1.1a)

	F⊣G	F's type	monad FG	1⊑FG	GF⊑1	F preserves ∪	G preserves ∩	FGF=F	GFG=G
1	◁⊣▷	A→ A⊗A	◁▷=𝟙	𝟙⊑◁▷	▷◁⊑𝟙	(R ∪ S)◁= R◁ ∪ S◁	(R∩S)▷= R▷∩S▷	◁▷◁=◁	▷◁▷=▷
2	⊸⊣⊸	A→𝕀	⊸⊸=⊤	𝟙⊑⊸⊸	⊸⊸⊑𝟙	(R ∪ S)⊸= R⊸ ∪ S⊸	(R∩S)⊸= R⊸∩S⊸	⊸⊸⊸=⊸	⊸⊸⊸=⊸
3	°⊣°	(A→B)→ (B→A)	R°° = R	R°° = R	R°° = R	(R ∪ S)° = S° ∪ R°	(R ∩ S)° = R° ∩ S°	R°° = R	R°° = R
4	⊸◁⊣▷⊸	(X⊗A→Y)→ (X→Y⊗A)	𝟙	(⊸◁⊗𝟙) (𝟙⊗▷⊸)=𝟙	(𝟙⊗⊸◁) (▷⊸⊗𝟙)=𝟙	=	=	=	=
5	▷⊸⊣⊸◁	(X→Y⊗A)→ (X⊗A→Y)	𝟙	(𝟙⊗⊸◁) (▷⊸⊗𝟙)=𝟙	(⊸◁⊗𝟙) (𝟙⊗▷⊸)=𝟙	=	=	=	=
6	Δ⊣∩	(A→B)→ (A→B)²	R ∩ R = R	R ∩ R = R	R ∩ S ⊑ R	Δ(R ∪ S)= ΔR ∪ ΔS	(R∩T)∩(S∩U)= (R∩S)∩(T∩U)	R ∩ R = R	R ∩ R = R
7	∪⊣Δ	(A→B)²→ (A→B)	(R,S)↦ (R ∪ S,R ∪ S)	R ⊑ R ∪ S	R ∪ R = R	(R ∪ T) ∪ (S ∪ U)= (R ∪ S) ∪ (T ∪ U)	Δ(R∩S)= ΔR∩ΔS	R ∪ R = R	R ∪ R = R
8	𝟘⊣!	{*}→ (A→B)			𝟘 ⊑ R				
9	·S⊣/S	(A→B)→ (A→C)	S/S	R⊑(RS)/S	(T/S)S⊑T	(S ∪ T) R = (S R) ∪ (T R)	(R ∩ R') / S = (R / S) ∩ (R' / S)	((RS)/S)S =RS	((T/S)S)/S =T/S
10	S·⊣S\	(B→C)→ (A→C)	S\S	R⊑S\(SR)	S(S\T)⊑T	R (S ∪ T) = (R S) ∪ (R T)	S \ (R ∩ R') = (S \ R) ∩ (S \ R')	S(S\(SR)) =SR	S\(S(S\T)) =S\T
11	R∩⊣R⇒	(A→B)→ (A→B)	X↦R⇒(X∩R)	X⊑R⇒(X∩R)	R∩(R⇒Y)⊑Y	R∩(X ∪ Y)= (R∩X) ∪ (R∩Y)	R ⇨ S ∩ T = (R ⇨ S) ∩ (R ⇨ T)	R∩(R⇒(X∩R)) =X∩R	R⇒(R∩(R⇒Y)) =R⇒Y
12	𝒟⊣·⊤	(A→B)→ Cor A	R↦(𝒟R)⊤	R⊑(𝒟R)⊤	𝒟(A⊤)⊑A	𝒟(R ∪ S)= 𝒟R ∪ 𝒟S	(A∩B)⊤= A⊤∩B⊤	𝒟((𝒟R)⊤) =𝒟R	(𝒟(A⊤))⊤ =A⊤
13	ℛ⊣⊤·	(A→B)→ Cor B	R↦⊤(ℛR)	R⊑⊤(ℛR)	ℛ(⊤A)⊑A	ℛ(R ∪ S)= ℛR ∪ ℛS	⊤(A∩B)= ⊤A∩⊤B	ℛ(⊤(ℛR)) =ℛR	⊤(ℛ(⊤A)) =⊤A
14	·f⊣·f°	(A→B)→ (A→C)	ff°	Map(f) ⇒ 𝟙 ⊑ f f°	f°f⊑𝟙	(S ∪ T) R = (S R) ∪ (T R)	Simple(g) ⇒ (X ∩ Y) g° = (X g°) ∩ (Y g°)	ff°f=f	f°ff°=f°
15	f°·⊣f·	(A→C)→ (B→C)	ff°	Map(f) ⇒ 𝟙 ⊑ f f°	f°f⊑𝟙	R (S ∪ T) = (R S) ∪ (R T)	f(X∩Y)= fX∩fY	ff°f=f	f°ff°=f°
16	i⊣E	Map↪Rel	E	$\frac{𝟙}{∋}$:A→PA	∋:PB→B	—	—	$\frac{∋}{∋}$=𝟙	$\frac{R}{∋}$∋=R
17	·∋⊣ $\frac{}{∋}$	Map(A,PB)→ (A→B)	𝟙	$\frac{f∋}{∋}$=f	$\frac{R}{∋}$∋=R	—	—	$\frac{f∋}{∋}$∋ =f∋	$\frac{\frac{R}{∋}∋}{∋}$ =$\frac{R}{∋}$
18	$\frac{}{∋}$ ⊣·∋	(A→B)→ Map(A,PB)	𝟙	$\frac{R}{∋}$∋=R	$\frac{f∋}{∋}$=f	—	—	$\frac{\frac{R}{∋}∋}{∋}$ =$\frac{R}{∋}$	$\frac{f∋}{∋}$∋ =f∋

§1.2. Composing adjunctions

(1.2a)




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Map(f) ⇒ f (R / S) = (f R) / S
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(1.2b)

		·T	T·	·g	g°·	T∩	°	⊶◁	Δ
1	**·S**	X / (S₁ S₂) = X / S₂ / S₁	S \ (X / W) = (S \ X) / W	R/(Sg)= (Rg°)/S	Map(f) ⇒ f (R / S) = (f R) / S	—	(Y/S)°= S°\(Y°)	(RS)^= R^(S⊗𝟙)	(R ∩ R') / S = (R / S) ∩ (R' / S)
2	**S·**	S \ (X / W) = (S \ X) / W	(S₁ S₂) \ X = S₂ \ (S₁ \ X)	S\(Rg°)= (S\R)g°	(g°S)\R= S\(gR)	—	(S\Y)°= Y°/S°	((S⊗𝟙)R)^= SR^	S \ (R ∩ R') = (S \ R) ∩ (S \ R')
3	**·f**	Map(f) ⇒ R / (f S) = (R / S) f°	T\(Rf°)= (T\R)f°	(R S)° = S° R°			(R S)° = S° R°		Simple(g) ⇒ (X ∩ Y) g° = (X g°) ∩ (Y g°)
4	**f°·**	Map(f) ⇒ f (R / S) = (f R) / S	(Tf°)\R= f(T\R)	—	(R S)° = S° R°		(R S)° = S° R°		f(T₁∩T₂)= fT₁∩fT₂
5	**R∩**	—	—	—	—	R ⇨ S ⇨ T = R ∩ S ⇨ T	(R⇒Y)°= R°⇒(Y°)	—	R ⇨ S ∩ T = (R ⇨ S) ∩ (R ⇨ T)
6	**°**	(Y/T)°= T°\(Y°)	(T\Y)°= Y°/T°	(R S)° = S° R°	(R S)° = S° R°	(T⇒Y)°= T°⇒(Y°)	R°° = R	(R^)°= (R°⊗𝟙) (𝟙⊗▷⊸) both bends: the ° section's display	(T₁∩T₂)°= T₁°∩T₂°
7	**⊶◁**	(RT)^= R^(T⊗𝟙)	((T⊗𝟙)R)^= TR^	—	—	—	(R^)°= (R°⊗𝟙) (𝟙⊗▷⊸) both bends: the ° section's display	—	—
8	**∪**	(S ∪ T) R = (S R) ∪ (T R)	R (S ∪ T) = (R S) ∪ (R T)	(S ∪ T) R = (S R) ∪ (T R)	R (S ∪ T) = (R S) ∪ (R T)	R ∩ (S ∪ T) = R ∩ S ∪ R ∩ T	(R ∪ S)° = S° ∪ R°		
9	**0**	0 R = 0	R 0 = 0	0 R = 0	R 0 = 0	R ∩ 0 = 0	0° = 0		

§1.3. Adjoint triples

Beyond `∪⊣Δ⊣∩`, which §Section 1.1. already carries as two rows, the table holds one adjoint triple with content: `∃_f⊣f*⊣∀_f` along a map `f`, once on each side of the composite. `f : A→B` throughout this subsection.

(1.3a)

```
·f⊣·f°⊣/f°
f°·⊣f·⊣f\
```

The first two links of each chain are the map shunting rules, and the third is §Section 1.1.'s division row read at a chosen divisor: `·S⊣/S` at `S:=f°`, and `S·⊣S\` at `S:=f`.

The third link is not the first over again. For a map `f`, `/f` collapses to `·f°`; being a map is exactly what that collapse spends `Map(f) ⇒ R / (f S) = (R / S) f°` — `f°` is not a map, so `/f°` does not collapse in turn, and the three operators stay distinct. On `Rel(𝕀,A)=PA` they are the image triple:

(1.3b)

	operator	acts by	name
1	`·f`	`S↦f[S]`	direct image, `∃_f`
2	`·f°`	`T↦f⁻¹[T]`	inverse image, `f*`
3	`/f°`	`S↦{b : f⁻¹(b)⊆S}`	`∀_f`

§Section 1.1.'s `𝒟⊣·⊤` is this same chain along the projection `A⊗B→A`, read through `Rel(A,B)=P(A⊗B)`: `𝒟R={(a,a) : ∃b. aRb}`, `A⊤` is the relation that ignores `b` altogether, and the third link is `R↦𝟙∩R/⊤={(a,a) : ∀b. aRb}`.

(1.3c)

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𝒟⊣·⊤⊣𝟙∩·/⊤
```

Three is where it stops, and one map breaks both ends. Take `A={a₁,a₂}` and `B={b}`: `f[{a₁}]∩f[{a₂}]={b}` while `f[{a₁}∩{a₂}]=∅`, so `·f` does not preserve meets and has no left adjoint, and the same two subsets give `∀_f({a₁} ∪ {a₂})={b}` against `∀_f{a₁} ∪ ∀_f{a₂}=∅`, so `/f°` does not preserve joins and has no right adjoint. A monic `f` restores the binary case and no more — `⊤f` still reaches only `im f`, and `∀_f𝟘=B\im f` is still not `𝟘`. The chain extends only when `f°` is a map as well, and then `/f°=·f°°=·f` and it repeats forever. For an `S` with neither `S` nor `S°` a map there is no triple at all: `·S⊣/S` is two links and stops.

The rows that chain forever say nothing by it: `°` is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and `°⊣°⊣°` has no end, and §Section 1.1.'s other row of that kind, `⊶◁⊣▷⊶`, goes the same way.